SUNWAY UNIVERSITY

SCHOOL OF ENGINEERING AND TECHNOLOGY

COURSEWORK FOR

BSC (HONS) IN COMPUTER SCIENCE
BSC (HONS) INFORMATION TECHNOLOGY
BACHELOR OF SOFTWARE ENGINEERING (HONS)
BSC (HONS) INFORMATION SYSTEMS (DATA ANALYTICS)
BSC (HONS) INFORMATION TECHNOLOGY (COMPUTER NETWORKING AND SECURITY)
YEAR 1; ACADEMIC SESSION APRIL 2025

SEG1201: DATABASE FUNDAMENTALS

Answer Template and SQL Script and Video Presentation Submission: Due date :Week 14 TUESDAY (5 AUGUST 2025), 11.59 pm

SEG1201: DATABASE FUNDAMENTALS

GROUP NAME: 12_4 Animal Shelter

MEMBER:

NO	STUDENT NAME	STUDENT ID	STUDENT EMAIL
1	Sin Shen Nee	23050024	23050024@imail.sunway.edu.my
2	Ding Yee Qing	23080732	23080732@imail.sunway.edu.my
3	Lim Shan En	23080690	23080690@imail.sunway.edu.my
4	Lee Hui Wen	23079692	23079692@imail.sunway.edu.my
5	Tee Yu Qing	23077787	23077787@imail.sunway.edu.my

YOUTUBE LINK: https://youtu.be/PGrHPvCoPJU

INSTRUCTIONS

- This final assessment contributes 50% to your final grade.
- This assignment is primarily for Course Learning Outcome 2 - Implement a database design group project using appropriate tools such as Oracle SQL.

IMPORTANT

The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

Academic Honesty Acknowledgement

"We Sin Shen Nee, Ding Yee Qing, Lim Shan En, Lee Hui Wen, Tee Yu Qing**,** verify that this paper contains entirely our own work. We have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, we have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. We realize the penalties *(refer to page 16, 5.5, Appendix 2, page 44 of the student handbook diploma and undergraduate programme)* for any kind of copying or collaboration on any assignment."

YUQING, HUIWEN, SHENNEE, SHANEN, YEEQING

3/8/2025

(Student signatures/Date)

PART 1: GENERATE A CASE SCENARIO AND DESIGN A RELATIONAL DATA MODEL (35 MARKS)

Scenario

Paw Heaven Database System

Paw Heaven Animal Shelter is a non-profit charity that is located in the center of Cheras, Selangor. It was founded in 2015 with a simple and meaningful mission, "**Rescue, Recare and Rehome**." This means the shelter is dedicated to rescuing stray and abandoned animals by providing them with proper healthcare and medical attention, and helping them find a loving home through adoption processes. Over the years, Paw Heaven has grown and expanded its facilities and services to accommodate the increasing number of animals in need. However, the administration of records and operations has become increasingly disorganized and ineffective as more and more animals continue to arrive. In order to better manage information related to animals, the shelter creates a structured database system to facilitate daily operations such as caretaker scheduling, adoption management, healthcare monitoring and animal registration.

The shelter accepts animals from rescue groups, individuals who drop them off, or transfers from other organizations. First, each **animal** that comes to the shelter is assigned a unique identification number and some basic information such as name, gender, date of birth (if known), and arrival date. The shelter houses a variety of **species**, including dogs, cats, rabbits, and birds. To ensure that animals are put in the right groups, each one is assigned to a species category. This helps caretakers and vets give the appropriate meals and care routines.

Caretakers are an important part of the shelter's daily routine. The database stores each caretaker's first and last name, contact number, and shift time so that scheduling and management can be done more quickly. To ensure animals always get the same care, the shelter enforces a rule which states that each animal must be assigned to at least one caretaker on every shift. This makes sure that no animal is alone and provides caretakers with the chance to do duties including feeding, cleaning, and checking on the animals' health. Additionally, each caretaker is in charge of more than one animal during each shift. This helps to maximize the best use of staff allocation and shelter resources. If caretakers notice any sign of illness, they should tell the veterinary immediately so they can take action on it.

Adoption service is also a main goal of Paw Heaven. Only individuals who are sure they want to adopt can sign up as adopters. This guarantees that every registration leads to

at least one confirmed adoption. If an adopter is approved, they can adopt one or more animals. If they want to, they can return later to adopt more animals. The system keeps track of adopter's personal information, including name, identification number, contact number, address, and adoption date to ensure they can be followed up. To encourage responsible ownership, each animal can be matched with only one adopter. This prevents shared responsibility, multiple claims, and the risks of abandonment. This method ensures that every adoption is meaningful, and that every animal has a good chance of living a better life.

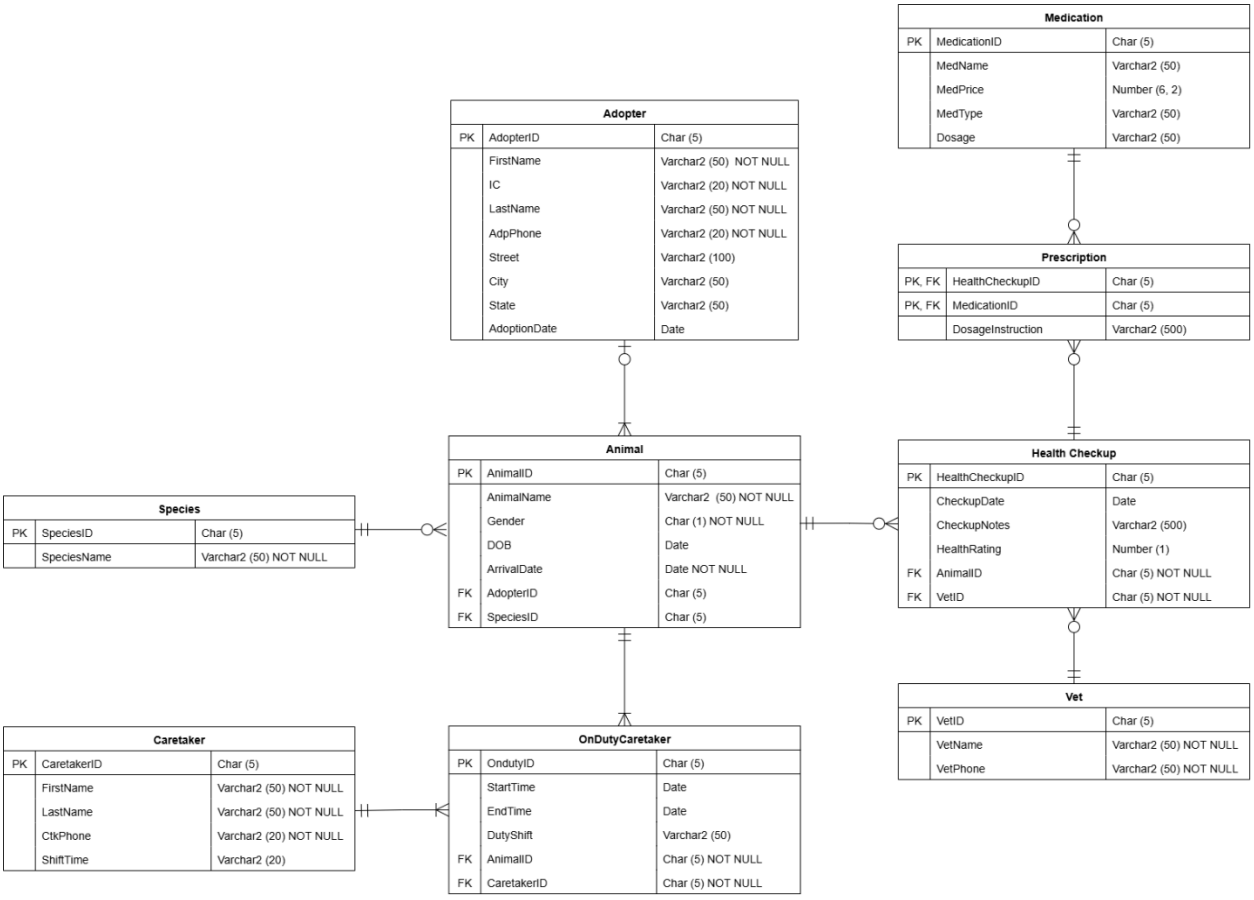
In addition, Paw Heaven maintains records of all the healthcare services offered to ensure the animals' well-being. The certified **veterinarians (Vets)** employed with the shelter conduct regular health checks on the animals. The database keeps records of each vet's name and contact number for future reference. Except for animals that have just arrived and haven't had a checkup yet, animals may have more than one health checkup while they are staying there. The long-term residents need to get these checkups, and new residents should have them as well. The system records a lot of information about each animal's health checkup, such as the checkup date, notes on the animal's condition, and any medications that were given. The **medications** records are stored by the shelter and sort them by name, type, and dosage instructions. This organized healthcare recording system helps the shelter monitor animal health and ensure they get the best care possible.

Through this database system, Paw Heaven Animal Shelter makes sure that all parts of its operations run smoothly and efficiently, from taking animals to classifying species, managing their health, and assigning caretakers. This system not only enhances the shelter's internal processes better, but it also builds trust and confidence as well as honesty with adopters, vets, and the community. By maintaining records in an orderly manner, the shelter can focus on its mission of providing animals with a safe and loving environment.

Business Rules

1. Each animal representative belongs to exactly one species.
2. Each species can have zero or many animals.
3. An animal may or may not be adopted.
4. Only individuals who intend to adopt can register as adopters. Every adopter must adopt at least one animal.
5. Each animal can have zero or many health checkups.
6. Each health checkup must be recorded for exactly one animal.
7. Each health checkup must be conducted by one vet.
8. Each vet can perform zero or more health checkups.
9. Each health checkup can involve zero or more medications.
10. Each medication can be used in zero or many health checkups.
11. During any shift, each animal must be assigned to at least one caretaker.
12. Each caretaker is assigned to multiple animals in each shift.

Entity Relationship Diagram



Constraints

Adopter Table Constraints

	Constraint	Justification
1	AdopterID LIKE 'A%'	Ensures that each AdopterID follows a consistent pattern, which starts with uppercase letter ‘A’ for easier identification and management.
2	IC UNIQUE	Ensures that each adopter’s IC number is unique to prevent duplicate entries and minimize typos that may cause identification issues.
3	IC NOT NULL	Ensures that each adopter provides an IC number for record completeness.
4	FirstName NOT NULL	Ensures that each adopter provides a first name for identification.
5	FirstName = UPPER(FirstName)	Enforces consistent formatting by recording every first name in uppercase.
6	LastName NOT NULL	Ensures that each adopter provides a last name for identification.
7	LastName = UPPER(LastName)	Enforces consistent formatting by recording every last name in uppercase.
8	AdpPhone NOT NULL	Ensure every adopter provides a valid phone number to enable follow-up communication regarding the adopted animal(s).
9	AdpPhone LIKE '0%-%'	Enforces adopter phone number with Malaysian phone number format (e.g., 012-3456789).
10	AdoptionDate DEFAULT SYSDATE	Records current adoption date automatically to maintain accuracy of record.

Species Table Constraints

	Constraint	Justification
1	SpeciesID LIKE 'S%'	Ensures that each SpeciesID follows a consistent pattern, which starts with uppercase letter ‘S’ for easier identification and management.
2	SpeciesName NOT NULL	Ensures that each species has a clear name to allow classification of animals.

Animal Table Constraints

	Constraint	Justification
1	AnimalID LIKE 'AN%'	Ensures that each AnimalID follows a consistent pattern, which starts with uppercase letters ‘AN’ for easier identification and management.
2	AnimalName NOT NULL	Ensure each animal has a name for identification.
3	Gender NOT NULL	Ensures that each animal has a gender recorded, which will be helpful in health care purposes.
4	Gender IN ('M', 'F')	Restricts gender values to valid codes; ‘M’ for male, ‘F’ for female to ensure consistency.
5	ArrivalDate NOT NULL	Ensures that the arrival date is stored when animal arrived at the shelter.
6	FOREIGN KEY (AdopterID)	Presents an optional relationship that an animal may or may not be adopted by an adopter.
7	FOREIGN KEY (SpeciesID)	Ensure each animal is assigned to a valid species.

Caretaker Table Constraints

	Constraint	Justification
1	CaretakerID LIKE 'C%'	Ensures that each CaretakerID follows a consistent pattern, which starts with uppercase letter ‘C’ for easier identification and management.
2	FirstName NOT NULL	Ensures that a first name is recorded to identify each caretaker.
3	LastName NOT NULL	Ensures that a last name is recorded to identify each caretaker.
4	CtkPhone NOT NULL	Ensures that a phone number is stored to reach out in work coordination and communication.
5	CtkPhone LIKE '0%-%'	Enforces phone number with Malaysian phone number format (e.g., 012-3456789).
6	CtkPhone UNIQUE	Prevents duplicate caretaker phone numbers to ensure each caretaker has a distinct contact.

On Duty Caretaker Table Constraints

	Constraint	Justification
1	OndutyID LIKE 'O%'	Ensures that each OndutyID follows a consistent pattern, which starts with uppercase letter ‘O’ for easier identification and management.
2	AnimalID NOT NULL	Ensures that each shift involves at least one animal which links with the requirement that every animal must be cared for.

3	CaretakerID NOT NULL	Ensures that each shift is assigned at least one caretaker.
4	FOREIGN KEY (AnimalID)	Links the on-duty record to a valid animal.
5	FOREIGN KEY (CaretakerID)	Links the shift record to a valid caretaker for scheduling and tracking.

Vet Table Constraints

	Constraint	Justification
1	VetID LIKE 'V%'	Ensures that each VetID follows a consistent pattern, which starts with uppercase letter 'V' for easier identification and management.
2	VetName NOT NULL	Ensures that each vet's name is recorded to track where conducted checkups.
3	VetPhone NOT NULL	Ensures that each vet phone number is stored for contact.
4	VetPhone LIKE '0%-%'	Enforces phone number with Malaysian phone number format (e.g., 012-3456789).
5	VetPhone UNIQUE	Prevents duplicate caretaker phone numbers to ensure each caretaker has a distinct contact.

Health Checkup Table Constraints

	Constraint	Justification
1	HealthCheckupID LIKE 'H%'	Ensures that each HealthCheckupID follows a consistent pattern, which starts with uppercase letter 'H' for identification and management.

2	HealthRating CHECK BETWEEN 1 AND 5	Restricts health rating to a standard range (1-5) to allows consistent evaluation of each animal health status.
3	AnimalID NOT NULL	Ensures that each health checkup is linked to one animal.
4	VetID NOT NULL	Ensure that each health checkup is conducted by a registered vet.
5	FOREIGN KEY (AnimalID)	Ensures that health checkups are only recorded for valid animals.
6	FOREIGN KEY (VetID)	Ensures that each health checkup is linked to the vet who performed it.

Medication Table Constraints

	Constraint	**Justification**
1	MedicationID LIKE 'M%'	Ensures that MedicationID follows a consistent pattern, which starts with uppercase letter ‘M’ for identification and management.
2	MedType IN ('Pill', 'Topical', 'Syrup', 'Injection', 'Tablet')	Restricts medication types input to defined into five categories to ensure consistent medication type data.

Prescription Table Constraints

	Constraint	**Justification**
1	PRIMARY KEY (HealthCheckupID, MedicationID)	Ensures that each prescription is specially linked from a medication to a health checkup.

2	FOREIGN KEY (HealthCheckupID)	Ensures that each prescription references a valid health checkup record.
3	FOREIGN KEY (MedicationID)	Ensures that each prescription references a valid medication entry.

PART 2: IMPLEMENT AND QUERY A DATABASE USING SQL (30 MARKS)

For part 2, submit your script with a .sql file extension. The script will include the creation of all the tables, population of all the tables and the required queries.

***You do not have to edit this part.*

PART 4: DEMONSTRATION (10 MARKS)

List each student's presentation here. Please update the following table accordingly.

No	ID	Name	Questions to present
i)	23080732	Ding Yee Qing	Use slides to present the scenario, goals, ERD, design choices.
ii)	23079692	Lee Hui Wen	Use slides with quick table views to show how our database system supports scenario (ERD + tables).
iii)	23050024	Sin Shen Nee	Run 2 queries (a & b), explain results and justification.
iv)	23077787	Tee Yu Qing	Run 2 queries (c & d), explain results and justification.
v)	23080690	Lim Shan En	Run final 2 queries (e & f), summarize system strengths, novelty, and conclude.